

RootNode - Goals

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1. Introduction

Technical instructions for installing, configuring, and operating the hardware and firmware are available in the **README.md**. A separate **Goals.pdf** contains the learning objectives.

The device is intentionally vulnerable and should only be used in an isolated, authorized training environment. The described techniques must not be applied to systems without explicit permission.

If you discover additional vulnerabilities that are not documented in this manual, please report them to Luke de Munk so they can be reviewed and, where appropriate, incorporated into the project documentation and training material.

2. Goals

A list of objectives provides a convenient way to track progress while exploring and exploiting the vulnerabilities in this project. Some objectives can be achieved in multiple ways, encouraging to investigate different attack paths. This chapter presents only the objectives. The exploitable vulnerabilities and the methods used to exploit them are described in the following chapter.

1. **Gain access to the power and brightness controls**
2. **Gain access to the serial CLI**
3. **Log in to the web interface as a regular user**
4. **Gain administrator access to the web interface**
5. **Download files containing sensitive information**
6. **Gain root privileges in the serial CLI**
7. **Obtain the credentials of the configured web users**
8. **Modify the web-user configuration by uploading a manipulated file**
9. **Automate the control of multiple devices on the same network**
10. **Install or execute unauthorized firmware on the device**
11. **Buffer overflow**